

EXPERIMENT 18

CI/CD PIPELINE USING JENKINS AND DOCKER

Aim: To set up a Jenkins CI/CD pipeline that automatically builds, tests, and deploys a containerized application, with Jenkins builds triggered by code commits or pull requests.

Software Requirements

- Jenkins
- Git and GitHub
- Docker
- A sample application
- Dockerfile
- Jenkinsfile
- Cloud platform such as AWS or GCP

Procedure

- 1. Set up Jenkins:** Install Jenkins on a local machine or server and open the Jenkins dashboard.
- 2. Create the application:** Prepare a simple sample application with the required source files and tests.
- 3. Create the Dockerfile:** Define the base image, copy application files, install dependencies, expose the required port, and specify the application start command.
- 4. Create the Jenkinsfile:** Define pipeline stages for source checkout, Docker image build, testing, and deployment.
- 5. Configure the Jenkins Job:** Connect Jenkins to the GitHub repository and configure the job to use the Jenkinsfile from source control.
- 6. Configure automatic triggers:** Enable webhook or SCM polling so that commits or pull requests can trigger a new Jenkins build.
- 7. Build and test:** Jenkins checks out the latest source code, builds the Docker image, and executes the application tests.
- 8. Deploy:** After successful testing, push the image to a container registry and deploy the containerized application to the selected cloud platform.
- 9. Verify the pipeline:** Check the Jenkins stages and console output to confirm that all stages completed successfully.

Sample Dockerfile

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]
```

Sample Jenkinsfile

```
pipeline {
  agent any
  stages {
    stage('Checkout') { steps { checkout scm } }
    stage('Build') { steps { sh 'docker build -t sample-app:latest .' } }
    stage('Test') { steps { sh 'docker run --rm sample-app:latest pytest' } }
    stage('Deploy') { steps { echo 'Deploy to AWS/GCP' } }
  }
}
```

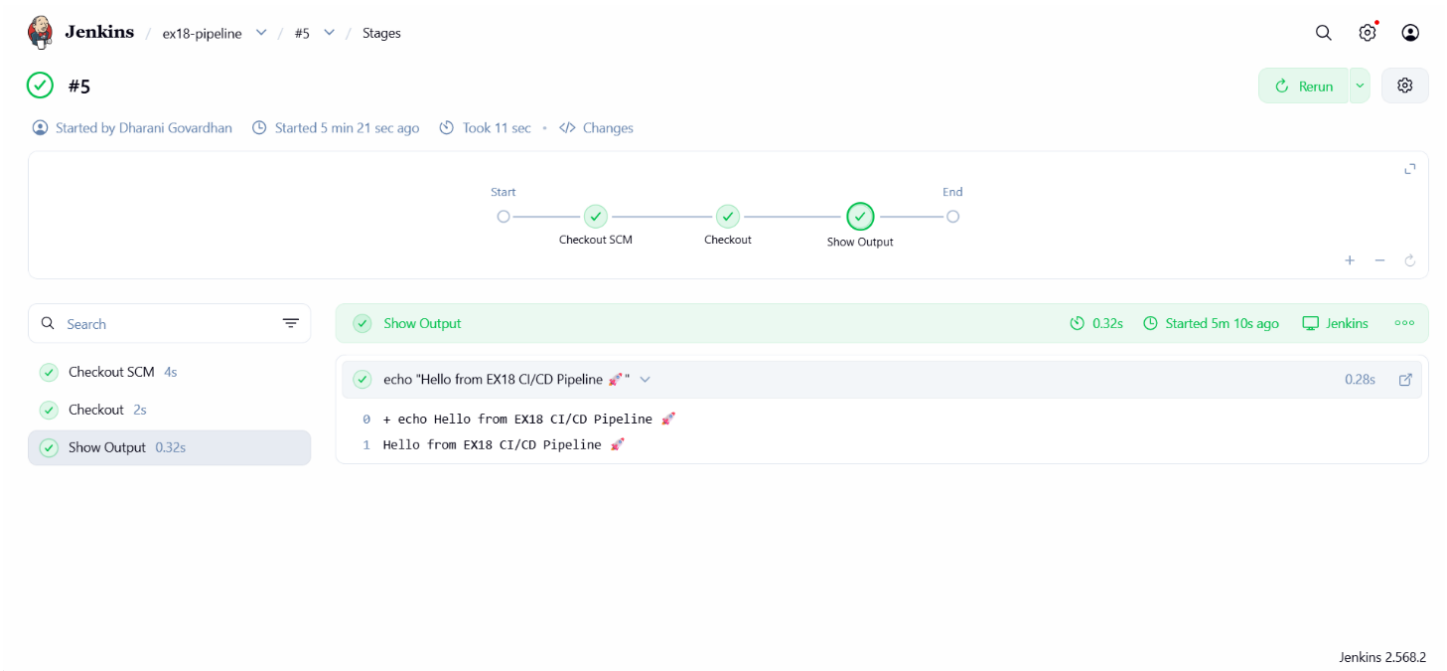
Pipeline Stages

- Checkout SCM – retrieves the latest source code.
- Build – creates the Docker image.
- Test – runs automated tests on the application.
- Deploy – publishes/deployes the successful build to the cloud environment.
- Trigger – starts the pipeline after a commit or pull request.

Result

The Jenkins CI/CD pipeline was configured successfully. The screenshot shows a successful Jenkins build with completed Checkout SCM, Checkout, and Show Output stages. The console output confirms the pipeline executed the sample message successfully.

SCREENSHOT OF COMPLETED JENKINS PIPELINE



Conclusion: A CI/CD workflow using Jenkins was demonstrated. The pipeline automates source checkout, application build, testing, and deployment preparation, providing a repeatable workflow for containerized applications.